

Objectives

At the end of this topic, students are expected to be able to:

1. expand the expression $(x + y)^n$.
2. state and prove binomial theorem for a positive integral index.
3. state without proof binomial theorem when n is not a positive integer.
4. apply binomial theorem to solve real-life problems.

Introduction

Let us consider the expansion of the following two expressions of the general form $(x + y)^n$.

$$\begin{aligned} 1. (1 + x)^2 &= (1 + x)(1 + x) \\ &= 1 + 2x + x^2 \end{aligned}$$

$$\begin{aligned} 2. (1 + x)^3 &= (1 + x)(1 + x)(1 + x) \\ &= 1 + 3x + 3x^2 + x^3 \end{aligned}$$

However, when we try to expand $(1 + x)^4$ and $(1 + x)^5$, we find that this method become very tedious. Furthermore, how about $(1 + x)^n$ when n is an arbitrary positive integer? The binomial theorem helps us to expand these and other expressions more easily and conveniently.

The Binomial Coefficients

Definition: (Binomial Coefficient)

The binomial coefficients are the coefficients in the expansion of $(1 + x)^n$. The coefficient of x^r in $(1 + x)^n$ is denoted as ${}_nC_r$ or $\binom{n}{r}$.

(Here, we shall be using the later notation throughout.)

Pascal's Triangle

Pascal's triangle is named after the French mathematician, Blaise Pascal (1623 – 1662), although it had been described 5 centuries earlier by the Chinese and also by the Persians. The

binomial expansion was known for the case $n = 2$ by Euclid ca 300 BC, and stated in modern form by Pascal.

Recall that:

$(1 + x)$	=	$1 + x$
$(1 + x)^2 = (1 + x)(1 + x)$	=	$1 + 2x + x^2$
$(1 + x)^3 = (1 + x)(1 + x)(1 + x)$	=	$1 + 3x + 3x^2 + x^3$
$(1 + x)^4 = (1 + x)(1 + x)(1 + x)(1 + x)$	=	$1 + 4x + 6x^2 + 4x^3 + x^4$
$(1 + x)^5 = (1 + x)(1 + x)(1 + x)(1 + x)(1 + x)$	=	$1 + 5x + 10x^2 + 10x^3 + 5x^4 + x^5$

Fig.1

and so on.

From the expansions, it is observed that

- i. the first and last coefficient in each expansions is 1.
- ii. every other number in the coefficient in a particular expansion can be obtained by adding the two coefficients directly above it.
- iii. the expansion from left to right and vice versa mirror symmetry
- iv. the coefficients in each row sum to twice the value of the sum of coefficients of the previous row.

Following this pattern, we obtained what is known as Pascal's triangle:

0th row	1					
1st row		1		1		
2nd row		1	2	1		
3rd row		1	3	3	1	
4th row	1	4	6	4	1	
5th row	1	5	10	10	5	1

Fig.2

The 0th row is the coefficient in the expansion of $(1 + x)^0$, or simply 1. From the first row onwards, the n th row contains the $n + 1$ coefficients in the expansion of $(1 + x)^n$. As you can probably observe, there are some intriguing properties about Pascal's triangle:

1. the triangle is symmetric about the middle of each row, i.e. $\binom{n}{r} = \binom{n}{n-r}$.

2. except for the leftmost and rightmost numbers in each row, every number is the sum of the two numbers directly above it, i.e. $\binom{n+1}{r} = \binom{n}{r-1} + \binom{n}{r}$.

3. In every row, the numbers first increase from 1 to a maximum value, then decrease back to 1.

4. the second number (as well as the second last number) in the n th row is n ,

$$\text{i.e. } \binom{n}{1} = \binom{n}{n-1} = n$$

Exercise 1. Prove that $\binom{n+1}{r} = \binom{n}{r-1} + \binom{n}{r}$.

In order to generalize the expansion of $(1+x)^n$ in powers of x , it is easily observed from the few expansions we have, that the coefficient of x^r in the expansion is $\binom{n}{r}$, where

$$\begin{aligned} \binom{n}{r} &= \frac{n(n-1)(n-2) \dots (n-r+1)}{r(r-1)(r-2) \dots 3.2.1} \\ &= \frac{n!}{r!(n-r)!} \end{aligned}$$

For instance,

$$\begin{aligned} \binom{5}{2} &= \frac{5(5-1)(5-2)!}{2!(5-2)!} \\ &= \frac{5(5-1)}{2(2-1)} = \frac{5.4}{2.1} \end{aligned}$$

Thus,

$$(1+x)^n = 1 + \binom{n}{1}x + \binom{n}{2}x^2 + \dots + \binom{n}{r}x^r + \dots + \binom{n}{n-1}x^{n-1} + x^n \quad (1)$$

Note from 1. that $\binom{n}{n} = \binom{n}{0} = 1$

Let us assume that (1) is valid, then

$$\begin{aligned} (a+x)^n &= \left[a \left(1 + \frac{x}{a} \right) \right]^n = a^n \left[1 + \binom{n}{1} \frac{x}{a} + \binom{n}{2} \frac{x^2}{a^2} + \dots + \binom{n}{r} \frac{x^r}{a^r} + \dots + \binom{n}{n-1} \frac{x^{n-1}}{a^{n-1}} + \frac{x^n}{a^n} \right] \end{aligned}$$

Therefore,

$$(a + x)^n = a^n + \binom{n}{1} x a^{n-1} + \binom{n}{2} x^2 a^{n-2} - \dots + \binom{n}{r} x^r a^{n-r} + \dots + x^n \quad (2)$$

Thus, (1) and (2) respectively can be written using notation as:

$$(1 + x)^n = \sum_{r=0}^n \binom{n}{r} x^r \quad (3)$$

$$(a + x)^n = \sum_{r=0}^n \binom{n}{r} x^r a^{n-r} \quad (4)$$

Example 1: Using the Pascal's triangle, expand $(1 + x)^5$.

Solution:

From the Pascal's triangle the terms in the 5th row are 1, 5, 10, 10, 5 and 1.

Thus we have $(1 + x)^5 = x^5 + 5x^4 + 10x^3 + 10x^2 + 5x + 1$

Example 2: Use the Pascal's triangle to expand $(10 + x)^6$.

Solution: We shall discuss the solution in the class.

Exercise 2. Determine, using Pascal's triangle method, the expansion of $(2p - 3q)^5$.

The coefficients in the expansions of $(1 + x)^n$ in the figure above, is equivalent to the following combinations of n things k at a time:

$\binom{0}{0}$					
$\binom{1}{0}$		$\binom{1}{1}$			
$\binom{2}{0}$		$\binom{2}{1}$	$\binom{2}{2}$		
$\binom{3}{0}$	$\binom{3}{1}$		$\binom{3}{2}$	$\binom{3}{3}$	
$\binom{4}{0}$	$\binom{4}{1}$	$\binom{4}{2}$	$\binom{4}{3}$	$\binom{4}{4}$	

Fig.3

The Binomial Theorem

We are going to find an explicit formula for computing the numbers in the Pascal's triangle, so that we can carry out the expansions conveniently without always referring to the triangle as well as explaining the interesting properties we observed.

According to definition,

$$(1+x)^n = (1+x)^n = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \dots + \binom{n}{n-1}x^{n-1} + \binom{n}{n}x^n \quad (5)$$

Putting $x = 0$ in (5) we get $\binom{n}{0} = 1$ as we can see from the Pascal's triangle. The other terms can also be computed by successive differentiation of (5) above as illustrated below.

Differentiating (5) once, we have:

$$n(1+x)^{n-1} = \binom{n}{1} + 2\binom{n}{2}x + \dots + (n-1)\binom{n}{n-1}x^{n-2} + n\binom{n}{n}x^{n-1} \quad (6)$$

Putting $x = 0$ again in (6), we have

$$\binom{n}{1} = n.$$

Similarly, by differentiating (5) r times, we have

$$n(n-1)(n-2)\dots(n-r+1)(1+x)^{n-r} = r\dots 3.2.1\binom{n}{r} + (r+1)r\dots 3.2\binom{n}{r+1}x + \dots + n(n-1)(n-2)\dots(n-r+1)\binom{n}{n}x^{n-r} \quad (7)$$

Putting $x = 0$ in (7), we get

$$\binom{n}{r} = \frac{n(n-1)(n-2)\dots(n-r+1)}{r\dots 3.2.1} \quad (8)$$

To make the formula (8) simpler, we introduce the factorial symbol.

Definition: (Factorial)

For positive integer n , we define the factorial of n , denoted by $n!$ by

$$n! = n(n-1)(n-2)\dots 3.2.1$$

We also define $0! = 1$

With this notation in hand, (8) can be written in factorial form as:

$$\binom{n}{r} = \frac{n!}{r!(n-r)!} \quad (9)$$

Proof of Binomial Theorem when n is a Positive Integer

We shall state and prove binomial theorem for positive integral index.

Theorem 1.

For $n \in \mathbb{N}$ and $a, x \in \mathbb{R}$, then

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k} \quad (10)$$

Proof

By induction on n . When $n = 0$ both sides evaluate to 1. Assume the claim holds for $n = m$ and consider the case $n = m + 1$.

$$(x + y)^{m+1} = (x + y)(x + y)^m$$

$$= (x + y) \sum_{k=0}^m \binom{m}{k} x^k y^{m-k}$$

$$= x \cdot \sum_{k=0}^m \binom{m}{k} x^k y^{m-k} + y \cdot \sum_{k=0}^m \binom{m}{k} x^k y^{m-k}$$

$$= \sum_{k=0}^m \binom{m}{k} x^{k+1} y^{m-k} + \sum_{k=0}^m \binom{m}{k} x^k y^{m+1-k} \quad (11)$$

$$= \sum_{k=1}^{m+1} \binom{m}{k-1} x^k y^{m+1-k} + \sum_{k=0}^m \binom{m}{k} x^k y^{m+1-k} \quad (12)$$

$$= x^{m+1} + \sum_{k=1}^m \binom{m}{k-1} x^k y^{m+1-k} + y^{m+1} + \sum_{k=1}^m \binom{m}{k} x^k y^{m+1-k}$$

$$= x^{m+1} + y^{m+1} + \sum_{k=1}^m \left[\binom{m}{k-1} + \binom{m}{k} \right] x^k y^{m+1-k} \quad (13)$$

$$= x^{m+1} + y^{m+1} + \sum_{k=1}^m \binom{m+1}{k} x^k y^{m+1-k} \quad (14)$$

$$= \sum_{k=1}^{m+1} \binom{m+1}{k} x^k y^{m+1-k} \quad \blacksquare \quad (15)$$

Here (12) follows from (11) by noting that

$$\sum_{k=0}^m f(k) = \sum_{k=1}^{m+1} f(k-1) \quad (16)$$

and (14) follows from (13) by Exercise 1 (Pascal's rule). The other steps are simple algebraic manipulation. This completes the proof by induction.

Note:

By substituting $x = y = 1$ into (10), we get the identity:

$$\sum_{k=0}^n \binom{n}{k} = 2^n \quad (17)$$

Example 3: Expand $(x + 2y)^7$. Hence evaluate $(1.02)^7$ to four significant figures.

Solution.

By (10),

$$\begin{aligned} (x + 2y)^7 &= \sum_{k=0}^7 \binom{7}{k} x^k (2y)^{7-k} \\ &= x^7 + 7x^6(2y) + 21x^5(2y)^2 + 35x^4(2y)^3 + 35x^3(2y)^4 + 21x^2(2y)^5 + 7x(2y)^6 + (2y)^7 \\ &= x^7 + 14x^6y + 84x^5y^2 + 280x^4y^3 + 560x^3y^4 + 672x^2y^5 + 448xy^6 + 128y^7 \end{aligned}$$

By re-writing $(1.02)^7$ as $(1 + 0.02)^7$ and compare with $(x + 2y)^7$, then with $x = 1$ and $y = 0.01$, we obtain

$$\begin{aligned} (1.02)^7 &= 1^7 + 14(0.01) + 84(0.01)^2 + 280(0.01)^3 + 560(0.01)^4 + 672(0.01)^5 \\ &\quad + 448(0.01)^6 + 128(0.01)^7 \\ &= 1.149 \text{ to four significant figures} \end{aligned}$$

Example 4: Find the term independent of y in the expansion of $\left(\frac{x^4}{y^3} + \frac{y^2}{2x}\right)^{10}$

Solution

$$\begin{aligned} \left(\frac{x^4}{y^3} + \frac{y^2}{2x}\right)^{10} &= \sum_{k=0}^{10} \binom{10}{k} \left(\frac{x^4}{y^3}\right)^k \left(\frac{y^2}{2x}\right)^{10-k} \\ &= \sum_{k=0}^{10} \binom{10}{k} 2^{k-10} x^{5k-10} y^{20-5k} \end{aligned}$$

Thus, the term independent of y will have y raised to zero; hence

$$k = 4.$$

Thus, the coefficient of the term is

$$\binom{10}{k} 2^{k-10}$$

$$\text{Hence, when } k = 4 \quad \binom{10}{k} 2^{k-10} = \binom{10}{4} 2^{4-10} = \frac{10!}{5!5!} 2^{-6}$$

$$= \frac{105}{32}$$

Hence, the term independent of y is $\frac{105}{32}x^{10}$

Example 5: Find the maximum coefficient in the expansion of $(3x + 5)^{10}$ without actual expansion.

Solution

Note that the k th term in the expansion is $\binom{10}{k} (3x)^k 5^{10-k}$

We consider the ratio of the $(k + 1)$ st term to the k th term, i.e.

$$\frac{\binom{10}{k+1} 3^{k+1} \cdot 5^{9-k}}{\binom{10}{k} 3^k \cdot 5^{10-k}} = \frac{3}{5} \cdot \frac{10-k}{k+1}$$

To see which coefficient is maximum, we only need to know when the ratio is greater than 1 and when it is smaller than 1. So we set

$$\frac{3}{5} \cdot \frac{10-k}{k+1} > 1$$

$$k > \frac{25}{8} = 3\frac{1}{8}$$

Therefore, the coefficient increase at first and reach a maximum when $k = 4$.

So, the maximum coefficient is $\binom{10}{4} 3^4 5^6 = 265781250$

Exercise 3. Write down the first four terms of the binomial expansion of $(1 - y)^8$ in ascending powers of y .

Exercise 4. By putting $y = \frac{1}{2}x(1 - x)$ in your expansion in Exercise 3, find the values of p and q if

$$\left(1 - \frac{1}{2}x(1 - x)\right)^8 = 1 - 4x = px^2 + qx^3 + \dots$$

Exercise 5. Find the coefficient of x^{22} in the expansion of $(1 - 3x)(1 + x^3)^{10}$

Exercise 6. Find the coefficient of a if the coefficient of x in the expansion of $(1 + ax)^8(1 + 3x)^4 - (1 + x)^3(1 + 2x)^4$ is zero. What is the coefficient of x^2 ?

Binomial Theorem when n is not a Positive Integer

Here, we shall state without prove, the binomial theorem when n is not a positive integer.

Theorem 2. Let $|x| < 1$ and $n \in (\mathbb{Z}^+)^c$, then

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \frac{n(n-1)(n-2)(n-3)}{4!}x^4 + \dots \quad (18)$$

We can therefore have the theorem as tool for binomial expansion when n is not a positive integer. Note also that (18) is valid for n , when n is a positive integer, as the coefficients after the coefficient of x^n are all zero. Hence, for this case, the constraint that $|x| < 1$ is not necessary.

Example 7: Expand $\frac{1}{(1+2x)^3}$ in ascending powers of x as far as the term in x^3 , using (18)

Solution

Here, $n = -3$ and x is replaced by $2x$. Thus, the resulting expansion will be valid for $|x| < 1$. Using (18),

$$\begin{aligned} (1+2x)^{-3} &\approx 1 - 3(2x) + \frac{-3(-3-1)}{2!}(2x)^2 + \frac{-3(-3-1)(-3-2)}{3!}(2x)^3 \\ &\approx 1 - 3(2x) + \frac{-3(-4)}{2!}(2x)^2 + \frac{-3(-4)(-5)}{3!}(2x)^3 \\ &\approx 1 - 6x + 24x^2 - 80x^3 \end{aligned}$$

Example 7: Obtain the first four terms of the expansion of $(1+3x)^{\frac{1}{3}}$ in ascending power of x . Hence evaluate $\sqrt[3]{1.03}$ to five significant figures.

Solution.

Here, $n = \frac{1}{3}$ and x is replaced by $3x$. Thus, the resulting expansion will be valid for $|x| < \frac{1}{3}$. Using (18),

$$\begin{aligned} (1+3x)^{\frac{1}{3}} &\approx 1 + \frac{1}{3}(3x) + \frac{\frac{1}{3}\left(\frac{1}{3}-1\right)}{2!}(3x)^2 + \frac{\frac{1}{3}\left(\frac{1}{3}-1\right)\left(\frac{1}{3}-2\right)}{3!}(3x)^3 \\ &\approx 1 + x - x^2 + \frac{5}{3}x^3 \end{aligned}$$

To evaluate $\sqrt[3]{1.03}$, let $3x = 0.03$, then $x = 0.01$ is still valid for the expansion in this example. Thus,

$$\begin{aligned} (1+3(0.01))^{\frac{1}{3}} &\approx 1 + 0.01 - (0.01)^2 + \frac{5}{3}(0.01)^3 \\ &\approx 1.009 \text{ to five significant figures.} \end{aligned}$$

Example 8: Show that $\sqrt{9+x^2} \approx 3 + \frac{1}{6}x^2 - \frac{1}{216}x^4$. For what values of x is the expansion valid.

Solution.
We have

$$\begin{aligned} \sqrt{9+x^2} &= (9+x^2)^{\frac{1}{2}} = \left[9\left(1+\frac{x^2}{9}\right)\right]^{\frac{1}{2}} \\ &= \sqrt{9}\left(1+\frac{x^2}{9}\right)^{\frac{1}{2}} \\ &= 3\left(1+\frac{x^2}{9}\right)^{\frac{1}{2}} \end{aligned}$$